

# Zelalem Fissiha Demssie

**Fullsatck Developer — Business Analyst — Senior Lecturer**

Ethiopian Public Service University

College of Computing and Digitalization

Department of Information Systems

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## Education

2025–2026 : **MSc. in Project Management, Florida University of Southeast (FUSE), Florida, United States of America (USA).**

**Status:** Ongoing; expected completion by 2026.

2024–2025 : **Higher Diploma Program in Teaching in Higher Education (HDP), Ambo University, Ethiopia.**

**Grade:** Successfully Completed (Certificate Awarded on October 09, 2025)

2019–2020 : **MSc. Degree in Software Engineering, Bahir Dar University, Bahir Dar Institute of Technology (BIT), Bahir Dar, Ethiopia.**

*Thesis Title:* Developing a Class-Imbalance Software Defect Prediction Model Using SVM and Machine Learning Algorithms.

**Grade:** Cumulative GPA: 3.79/4.00

2012–2016 : **BSc. Degree in Computer Science (CS), Ambo University, Ambo, Ethiopia.**

*Project Title:* Developing a Web-based Drug Store Management System for Ethiopian Red Cross Pharmacy.

**Grade:** Cumulative GPA: 3.76/4.00, with project result A+.

## Professional Experience

2018 - Now: **Senior Lecturer, Ethiopian Public Service University, Addis Ababa, Ethiopia.**
Responsible for curriculum development, research, and teaching a wide range of courses outlined in the curriculum of IS, computer science, AI, databases, and computer networks.

2013 - 2017: **Lecturer, Ambo University, Ambo, Ethiopia**
Taught computer science courses, including AI, database systems, software engineering, programming, and related topics, using practical and student-centered approaches. Conducted research, supervised student projects, and contributed to software development and curriculum activities. Participated in community service and technology projects addressing institutional and community needs.

2016 - 2017: **Full Stack Developer, Ambo University, Ambo, Ethiopia**
Develop full-stack web applications and institutional information systems across frontend and backend, design and implement APIs, manage database architecture, conduct testing and deployment, provide technical documentation, and support digital transformation initiatives.

2009 - 2010: **Digital Marketing Officer, Welkesa and Chilalo Media, Addis Ababa, Ethiopia**
Managed websites and blogs, created engaging digital content, and supported digital marketing and online communication. Contributed to website management, content publishing, social media communication, and digital marketing initiatives.

## Research and Publication

2021 - 2022: **Zelalem Fissiha Demssie**, Ramesh, S., Rajalakshmi, R., Dwivedi, J. N., Selvakanmani, S., Pant, B., Bharath Kumar, N. (2022). Optimization of LEACH Protocol in Wireless Sensor Network Using Machine Learning. Computational Intelligence and Neuroscience, 2022, Article 5393251. https://doi.org/10.1155/2022/5393251.

2019-2020: **Zelalem Fissiha Demssie**, Software Defect Prediction Model Using Filter Feature Selection and Support Vector Machine Algorithms on Highly Class Imbalanced Software Dataset. Article Manuscript, December 2020.

## Projects and Community Service

2025: **Integrated Lecture Lab Framework for Enhancing Practical Programming Skills using Machine Learning Algorithms, Lead Developer - 2025**
Designed and coordinated a university-led digital literacy training program to strengthen teachers' digital competencies, improve ICT integration in teaching, and promote effective technology-enabled learning in local high schools.

2024: **Software Digital Literacy and ICT Integration Training for High School Teachers, Project Coordinator - 2024.**
Designed and coordinated a university-led digital literacy training program to strengthen teachers' digital competencies, improve ICT integration in teaching, and promote effective technology-enabled learning in local high schools.

2023: **Machine Learning-Based Student Academic Performance Prediction System, Lead Developer, 2023, Ambo University.**
Developed a predictive machine learning model using academic data to identify key performance factors and enable data-driven academic advising, early identification of at-risk students, and timely interventions.

Training: **Certificate of Recognition for the contribution to community capacity building, Community Service Trainer, Ambo University.**
Provided practical training on fundamental computer skills to staff of the Ambo City Administration and Housing Office, supporting digital competency development and effective use of ICT in public service delivery.

## Professional Development and Training

September 2025: **Awarded official Certificate of Virtual Program Completion in September 2025.**
Completed a 7-month Peace Corps USA and Ambo University virtual program focused on classroom English teaching methodology and professional development.

February 2025: **Workshop on E-Learning and ICT Integration in Higher Education**. Mastered digital pedagogy and modern educational technology tools to enhance curriculum delivery.

December 2024: **Training on Data Analysis and Visualization Using Python in December 2024.**
Developed expertise in data manipulation, exploratory analysis, and visualization using key Python libraries and frameworks.

January 2023: **Seminar on Project Planning, Analysis, and Monitoring Tools, January 2023.**
Gained core competencies in Agile, project lifecycle management, performance tracking, and evaluation frameworks.

October 2021: **Awarded official Certificate of Pedagogical Training Trainee on October 9-14, 2021.**
Completed intensive pedagogical training covering higher education teaching methodologies, instructional planning, active learning, continuous assessment, and action research.

## Professional Role and Contributions

September 2026: **Information Systems Curriculum Development Team Leader — EPSU.**
Led and developed an outcome-based Information Systems curriculum based on national and international standards, successfully completed in September 2026.

June 15, 2026: **Curriculum Development Leader — Software Engineering and Intelligent Systems.**
Led and developed the unique outcome-based SEIS Curriculum based on national and international standards, 2026, Ethiopian Public Service University.

December 2013: **Curriculum Review Team Leader — 2013, Ambo University** Reviewed and developed the Harmonized Curriculum Computer Science Program aligned with academic standards.

2017 - 2018: **Quality Assurance and Auditing — Computer Science, Ambo University.**
Serving as the manager of the quality of course delivery according to the academic schedule, ensuring examinations covered core contents and competencies, and verifying that examination questions followed pedagogical standards.

2016 - 2018: **National Exit Exam Task Force Team Leader — Ambo University.** Led and coordinated the university task force responsible for planning, organizing, and supporting National Exit Examination activities, ensuring effective coordination, implementation, and compliance with examination requirements.

2014 - 2018: **Final Year Industrial Project Advisor — Computer Science, Ambo University.** Supervised and advised final-year students in planning, developing, testing, and documenting industry-based software projects, ensuring technical quality and successful project completion.

2014 - 2016: **Course Module Development — Computer Science, Ambo University.** Developed three comprehensive, curriculum-aligned course modules meeting academic standards: Design and Analysis of Algorithms (February 3, 2023), Fundamentals of Software Engineering (December 15, 2024), and Object-Oriented Software Engineering (September 2025).

## Professional Training Experience

2025: **Peace Corps USA-Ethiopia — Virtual Service English Program Participant**.
Completed a 7-month Peace Corps and Ambo University virtual program focused on classroom English academic teaching and professional development.

2025: **Artificial Intelligence Fundamentals Nanodegree – Udacity (October 12, 2025)**
www.udacity.com/certificate/e/bfd1a7f0-660b-11ef-a769-fbba549f5727.

2025: **Programming Fundamentals Nanodegree – Udacity (July 12, 2025)**
https://www.udacity.com/certificate/e/7599bf1a-62d6-11f0-aaf7-8ffa89a58fd3

2025: **Data Analysis Fundamentals Nanodegree – Udacity (September 2025)**
https://www.udacity.com/certificate/e/bfd1a7f0-660b-11ef-a769-fbba549f572b

2025: **Android Developer Fundamentals Nanodegree – Udacity (September 2025)**
https://www.udacity.com/certificate/e/bfd1a7f0-660b-11ef-a769-fbba549f572c

2025: **E-SHE: OEX101's How to Build in Open EdX Studio — offered by Ministry of Education. (September, 2025).**

2024: **Master Class for Teaching Online (Open edX)—EdPlus, ASU Online (May, 2024).**

2017: **Pedagogical Training Trainee — Awarded Official Certificate, Ambo University.**
Completed intensive pedagogical training covering higher education teaching methodologies, instructional planning, active learning, continuous assessment, and action research.

2017: **Best Algorithm Design Award — Institute of Technology, Dire Dawa University.**
Awarded 1st place in the 2nd Annual Programming Week for designing an optimal "Bank Queuing System" algorithm (May 2017), Dire Dawa University.

## Research Interests and Experience

Research Focus: My research interests and experience include artificial intelligence, machine learning, deep learning, AI and data governance, data science, predictive analytics, educational data mining, AI for education, AI for public service delivery, software engineering, software quality assurance, algorithm design and optimization, wireless sensor networks, intelligent systems, and ICT for digital transformation. My research focuses on applying AI and data-driven approaches to intelligent systems, software development, network optimization, and education.

## Programming and AI Skills

Programming: **Python, JavaScript, TypeScript, Java, Kotlin, Swift, C++, C, R, Rust.**

Web Dev: **React, Angular, Vue.js, Node.js, Express.js, Flask, RESTful APIs, HTML5, CSS3.**

Software Eng: **Software Development Life Cycle, Requirements Engineering, Software Testing, Quality Assurance, Git, GitHub, Agile Development.**

Data Science: **Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Data Analysis, and Data Visualization.**

Databases: **MySQL, PostgreSQL, MongoDB, SQLite, Firebase, Apache, Oracle.**

ML Frameworks: **Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, Hugging Face, OpenCV.**

## Soft Skills

Communication: **Clear verbal and written expression across teams and stakeholders.**

Teamwork: **Strong ability to work productively in diverse group settings.**

Problem-Solving: **Analytical approach to identifying issues and delivering practical solutions.**

Adaptability: **Quick adjustment to changing priorities, environments, and challenges.**

Time Management: **Efficient prioritization and delivery of tasks under deadlines.**

## Language Skills

Proficiency:

| Language | Reading | Speaking | Writing | Listening |
|---|---|---|---|---|
| **Amharic** | Native | Native | Native | Native |
| **English** | Fluent | Fluent | Fluent | Fluent |
| **Afaan Oromo** | Good | Good | Good | Good |

## References

Available: **Professional and Academic References available upon request.**

Reference 1: **Mr. Habtamu Molla — Senior Lecturer, Department of IT.**
**Ethiopian Public Service University, Addis Ababa, Ethiopia.**
Phone: (+251) 975195143
Email: habtamumullu@gmail.com

Reference 2: **Mr. Getachew Worku — IT Project Manager.**
**Dashen Bank, Addis Ababa, Ethiopia.**
Phone: (+251) 918511867
Email: getachewworku34@gmail.com

Reference 3: **Dr. Warkew Legese — Research Cosultant and Director.**
**Ethiopian Public Service University, Addis Ababa, Ethiopia.**
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